

# Aldi Gratiang Panggabean

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## SUMMARY

Currently pursuing a Bachelor's degree in Electrical Engineering at Del Institute of Technology with a strong focus on Embedded Systems and IoT, I have built a solid foundation in robotics, automation, and electronic system development. My technical journey began at SMK Negeri 2 Siatas Barita, where I specialized in Industrial Electronics, gaining hands-on experience in residential electrical installations, WiFi network configurations, and computer lab setups through internships in electronics and IT. Passionate about IoT and robotics, I aspire to develop smart systems and automation technologies that drive innovation and efficiency. With proficiency in Robot Operating System (ROS), robotic programming, and embedded systems, I am highly adaptable, eager to solve complex engineering challenges, and committed to contributing to impactful technological advancements.

## EXPERIENCE

### Assistant Lecturer of Data Structures and Algorithms Course

Del Institute of Technology

September 2024 - December 2024, Laguboti, Indonesia

- Assisted in teaching second-year Electrical Engineering students fundamental concepts of data structures and algorithms.
- Conducted lab sessions and guided students in implementing data structures such as linked lists, stacks, queues, trees, and graphs using C.
- Provided hands-on guidance during lab sessions, helping students troubleshoot and solve programming challenges.
- Evaluated student performance through assessments and provided constructive feedback to support their learning progress.

### Student Laboratory Assistant of Electrical Circuit Course

Del Institute of Technology

September 2024 - December 2024, Laguboti, Indonesia

- Guided students in assembling, testing, and troubleshooting electrical circuits using laboratory equipment such as multimeters, oscilloscopes, and power supplies.
- Developing a comprehensive set of exercise problems aimed at improving students' comprehension of key concepts.
- Offering personalized guidance to students to ensure their understanding of fundamental subject matter.
- Facilitating tutorial sessions to support students who need extra assistance in mastering course material or completing assignments.

### Training Instructor – IoT for Smart Agriculture

Del Institute of Technology

April 2024 -May 2024, Laguboti, Indonesia

- Conducted hands-on training on IoT applications in agriculture for students and teachers.
- Introduced sensor-based monitoring systems using IoT technology to optimize agricultural productivity.
- Developed interactive learning modules on embedded systems, sensor integration, and cloud-based data analysis.

### Assistant Lecturer of Programming Laboratory Practice

Del Institute of Technology

January 2024 -May 2024, Laguboti, Indonesia

- Instructed and mentored second-semester Electrical Engineering students in the fundamentals of programming using C language.
- Developed and delivered comprehensive lesson plans and lab exercises to reinforce programming concepts and logic.
- Evaluated student performance through assessments and provided constructive feedback to support their learning progress.

### Junior Technician

Ariel Elektronik

July 2021 - December 2021, North Sumatra, Indonesia

- Conducted comprehensive electrical installations for residential properties, ensuring safety and functionality, resulting in a 20% increase in customer satisfaction ratings.
- Configured and installed WiFi routers for optimal network performance and connectivity.
- Designed and implemented computer lab setups for SMK N 2 Siatas Barita, resulting in improved access to technology resources for students and teachers.
- Configured the server for the ANBK exam, ensuring smooth administration of the exam for 11th-grade students at SMK N 2 Siatas Barita.
- Provided technical repair services for various electronic devices, achieving a 80% customer satisfaction rating.

## PROJECTS

### Dynamic PID Optimization for Line-Following Mobile Robot

January 2024 - Present

- Designed an adaptive PID control system to improve the autonomous navigation of a line-following mobile robot.
- Conducted experiments comparing dynamic PID tuning with traditional fixed-gain approaches, demonstrating improved responsiveness to varying track conditions.
- Implemented an optimization strategy to adjust PID parameters in real time, enhancing stability and performance in dynamic environments.

## **IoT-Based Temperature Monitoring System**

November 2024 - December 2024

- Developed an IoT-based temperature and humidity monitoring system as a final project for the Instrumentation Systems course.
- Implemented HTTP communication to transmit real-time sensor data to a web server, ensuring seamless data logging and visualization.
- Showcased skills in IoT development, web-based data visualization, and real-time alert systems, integrating both hardware and software components.

## **Smart Trash Bin With Mechanical Iris**

August 2024 - September 2024

- Designed and developed an automated smart trash bin with a mechanical iris mechanism for hands-free operation.
- Showcased expertise in microcontroller programming, sensor integration, and mechanical design.
- Integrated an ultrasonic sensor to detect approaching objects and trigger the opening mechanism.

## **Pelatihan Inovasi Bidang Pertanian Dengan Pemanfaatan Teknologi IoT di SMK Negeri 1 Sigumpar**

May 2024 - August 2024

- Conducted a training program on implementing IoT technology in agriculture at SMK Negeri 1 Sigumpar to enhance students' and teachers' technological skills.
- Addressed challenges in adopting IoT in agricultural education by providing theoretical and practical training sessions.
- Developed hands-on workshops to teach IoT-based smart farming solutions, improving productivity and efficiency in the agricultural sector.

## **Development of an IoT-Based Egg Incubator with Web Application**

April 2024 - July 2024

- Developed an IoT-based intelligent control system for an egg incubator, aimed at optimizing incubation conditions for increased hatch rates.
- Implemented automated monitoring and regulation of temperature, humidity, and egg turning frequency, ensuring ideal conditions.
- Designed and created a web application for real-time data monitoring and remote control, improving user accessibility and system management.

## **Design and Development of a Full-Wave Bridge Rectifier Power Supply**

April 2024 - May 2024

- Designed and developed a regulated power supply using a full-wave bridge rectifier to convert AC to DC efficiently.
- Implemented a rectification circuit using four diodes to achieve full-wave rectification, ensuring continuous power delivery.
- Documented circuit design, implementation, and performance analysis for academic and research purposes.

## **Development of a Hexapod Robot for Indonesian SAR Robot Contest 2024 Using Raspberry Pi and Inverse Kinematics**

Del Institute of Technology • February 2024 - July 2024

- Led the development of a hexapod robot for the Indonesian Search and Rescue (SAR) Robot Contest 2024, with a team of 5 members.
- Utilized Raspberry Pi as the main controller and implemented inverse kinematics to achieve precise leg movements.
- Designed and programmed the robot to autonomously navigate various terrains and obstacles, enhancing its efficiency in search and rescue missions.
- Achieved a 20% increase in task completion speed compared to previous designs, showcasing significant improvements in both hardware and software performance.

## **Implementation of a Smart Home Control System via Bluetooth HC-05 Using a Rule-Based Algorithm on the Arduino Uno**

Del Institute of Technology • October 2022 - November 2022

- Developed a smart home control system that allows users to wirelessly control home appliances using Bluetooth HC-05 and an Arduino Uno.
- Implemented a rule-based algorithm to automate device operations based on predefined conditions and user commands.
- Integrated multiple electronic components, including relays, sensors, and microcontrollers, for real-time control and monitoring.

## **Development of a 150W OCL Amplifier Using 2N3055**

SMK N 2 Siatas Barita • May 2022 - May 2022

- Designed and implemented a high-power, Output Capacitor-Less (OCL) audio amplifier capable of delivering 150W output.
- Successfully enhanced audio signal amplification, achieving louder and clearer sound output suitable for high-fidelity audio systems. This project involved detailed circuit design, component selection, and practical assembly, culminating in a robust and efficient audio amplifier.

## **Four-DOF Robotic Arm for Object Sorting Based on Color Recognition Using Adaptive Control**

SMK N 2 Siatas Barita • November 2021 - April 2022

- Developed a 4-degree-of-freedom (DOF) robotic arm for sorting blocks based on color.
- Utilized the TAOS TCS3200 color sensor to detect block colors and applied inverse kinematics for precise arm movement.
- Designed and programmed the control system to automate the sorting process, enhancing efficiency and accuracy

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## EDUCATION

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### IoT Engineer Camp Real Project

Indobot Academy • Depok, Indonesia • 2024

- The "IoT Engineer Camp Real Project" Independent Study Program at Indobot Academy is part of the MSIB (Certified Internship and Independent Study Program) under the Kampus Merdeka initiative. This program provides intensive training in Internet of Things (IoT), covering real IoT project development, hardware and software integration, and cloud-based system implementation.

### Bachelor of Electrical Engineering

Del Institute of Technology • Situluama Laguboti, Indonesia

- Currently in the fourth semester of the Bachelor of Electrical Engineering program at Del Institute of Technology. Excited to continue learning and gaining practical experience in the field of electrical engineering.

### Industrial Electronics

SMK Negeri 2 Siatas Barita • Siatas Barita, North Sumatra, Indonesia • 2022

- Achieved 3rd place in the Electronics category at the Provincial Vocational Student Skills Competition (LKS) in North Sumatra.
- Gained hands-on experience in electronics and practical applications through various projects and internships.

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## CERTIFICATIONS

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### IoT Engineer Camp Real Project - MSIB Batch 7

PT Ozami Inti Sinergi & Indobot Academy • 2025

- Successfully completed the Internet of Things (IoT) Engineer Camp Real Project as part of the Certified Independent Study Program (MSIB) Batch 7.
- Certified by PT. Ozami Inti Sinergi & Indobot Academy under the Kampus Merdeka - Ministry of Education, Indonesia initiative.

### Introduction to Basic Python Programming and It's Practical Application for Arduino

PT Tikitobot Robotika Indonesia • 2023

- This certificate confirms my expertise in Python, Successfully completed an intensive training program on Python programming fundamentals and its practical implementation with Arduino.

### JavaScript Course

Progate • 2023

- Successfully completed 7 study sessions covering fundamental JavaScript concepts for web development.

### 3rd Place – Provincial Vocational Skills Competition (LKS) in Electronics

Government of North Tapanuli Regency • 2022

- Recognized and awarded by the Government of North Tapanuli Regency and signed by Drs. Nikson Nababan, M.Si (Regent of North Tapanuli).

### 3rd Place Winner of Vocational Student Skills Competition in Electronics Field

North Sumatra Provincial Education Department • 2021

- This certificate confirms my expertise in electronics, including essential skills like microcontroller application and prototype design. It demonstrates my ability to solve problems and effectively communicate technical concepts, making me well-suited for roles in the electronics and technology industries.

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## ORGANISATIONAL EXPERIENCE

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### DEL ROBOTIC CLUB - Chairman

Del Institute of Technology

May 2024 - Present, Laguboti, Indonesia

- Organized robotics workshops, competitions, and exhibitions to enhance technical and teamwork skills among members.
- Managed resource allocation and project planning, ensuring the effective execution of club activities.
- Represented the club in institutional and national robotics events, promoting innovation and technological advancement.

### Interactive Workshop on Arduino UNO for High Schoolers - MC & Trainer

Del Institute of Technology

November 2024, Laguboti, Indonesia

- Initiated and led the Arduino Introduction Workshop for high school students in Toba and North Tapanuli districts.
- Coordinated logistics, materials, and speaker arrangements to ensure a smooth and engaging learning experience.
- Strengthened collaboration between the university and local high schools, fostering interest in STEM education.
- Successfully secured sponsorship support to sustain the second batch of training programs.

**Electrical Engineering Student Cadre Training - PIC**

**Del Institute of Technology**

**October 2024 , Laguboti, Indonesia**

- Mentoring and guiding 53 Electrical Engineering students throughout the cadre program, providing in-depth guidance for their personal and academic development.
- Delivering comprehensive explanations about electrical components and their functionalities, ensuring a profound understanding of technical aspects in the field of Electrical Engineering.

**DEL ROBOTIC CLUB - Member**

**Del Institute of Technology**

**January 2023 - May 2024, Laguboti, Indonesia**

- Contributed to the development and testing of robotic systems, including automation and sensor integration.
- Engaged in hands-on robotics projects, focusing on IoT, automation, and embedded systems.
- Participated in national and local robotics competitions, contributing to research, design, and development.

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**SKILLS**

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Hard Skills: Electronics, Electrical, Embedded System, Programming, Microcontroller Programming (ESP8266, ESP32, Raspberry Pi, STM32), HTML, CSS, C, JavaScript, Python, IoT, Robotics, ROS (Robot Operating System), Kinematics, Sensor Actuator, Electrical Wiring, Soldering Rework, Assembling, Prototype Design, Control Systems, Web Development for IoT, Data, Cloud Integration (Firebase, Thingspeak, Blynk), Computer Networking (Router Configuration, HTTP Requests, MQTT Protocol)

Soft Skills: Analytical, Circuit Analyst, Project Management, Planning, Communication, Presentation, Team Collaboration, Problem Solving, Technical Writing, Leadership, Workshop & Training Facilitation, Public Speaking, Research & Development, Innovation & Creativity

Tools: Arduino IDE, MySQL, Canva Design, Microsoft Office, Proteus, KiCad, SolidWorks, MQTT Tools & Brokers(Eclipse Mosquitto), Visual Studio Code, CV AVR, NI Multisim, DevC++